

Problem 14

Based on the general term of the series, a_n , conclude that the following series diverges:

$$\sum_{n=1}^{\infty} \sin(n\alpha) \quad (0 < \alpha < \pi).$$

Solution

Step 1: Identify the General Term

The general term of our series is:

$$a_n = \sin(n\alpha), \quad \text{where } 0 < \alpha < \pi.$$

Step 2: Analyze the Behavior of the General Term

For the series to converge, a necessary condition is that:

$$\lim_{n \rightarrow \infty} a_n = 0.$$

We need to determine whether $\lim_{n \rightarrow \infty} \sin(n\alpha)$ exists and equals zero.

Step 3: Show the Limit Does Not Exist or Is Not Zero

The sequence $\{\sin(n\alpha)\}$ does not converge to zero. To see this, note that:

Since $0 < \alpha < \pi$ and α is not a rational multiple of π (for most values), the sequence $\{n\alpha \pmod{2\pi}\}$ is dense in $[0, 2\pi]$ (by Weyl's equidistribution theorem). This means the values $\sin(n\alpha)$ oscillate and take on values throughout the interval $[-1, 1]$ infinitely often.

More directly, we can show that the limit does not exist:

Case 1: α is a rational multiple of π .

If $\alpha = \frac{p\pi}{q}$ where p, q are integers with $\gcd(p, q) = 1$ and $0 < \frac{p}{q} < 1$, then $\sin(n\alpha)$ is periodic with period dividing $2q$. The sequence will repeat values, and for most such α , it will not approach zero.

For example, if $\alpha = \frac{\pi}{2}$, then:

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & \text{if } n \equiv 0 \pmod{4} \\ 1 & \text{if } n \equiv 1 \pmod{4} \\ 0 & \text{if } n \equiv 2 \pmod{4} \\ -1 & \text{if } n \equiv 3 \pmod{4} \end{cases}$$

This sequence oscillates between $-1, 0, 1$ and does not converge to zero.

Case 2: α is an irrational multiple of π .

If α/π is irrational, the sequence $\{\sin(n\alpha)\}$ is equidistributed and takes on values throughout $[-1, 1]$. Therefore, the limit does not exist, and in particular, does not equal zero.

Step 4: Alternative Proof Using Subsequences

We can construct subsequences that converge to different values:

Since $0 < \alpha < \pi$, we have $\sin(\alpha) > 0$. For any $\epsilon > 0$, there exist infinitely many values of n such that $n\alpha$ is close to $\frac{\pi}{2} \pmod{2\pi}$, giving $\sin(n\alpha) \approx 1$.

Similarly, there exist infinitely many values of n such that $n\alpha$ is close to $\frac{3\pi}{2} \pmod{2\pi}$, giving $\sin(n\alpha) \approx -1$.

Since we can find subsequences converging to different values (or at least bounded away from zero), the sequence $\{\sin(n\alpha)\}$ does not converge to zero.

Step 5: Apply the Divergence Test

Since $\lim_{n \rightarrow \infty} \sin(n\alpha)$ either does not exist or is not equal to zero (for $0 < \alpha < \pi$), by the Divergence Test, the series:

$$\sum_{n=1}^{\infty} \sin(n\alpha)$$

diverges.

Final Answer

Conclusion

The series $\sum_{n=1}^{\infty} \sin(n\alpha)$ for $0 < \alpha < \pi$ **diverges** because:

$\lim_{n \rightarrow \infty} \sin(n\alpha)$ does not equal zero (or does not exist).
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The sequence $\{\sin(n\alpha)\}$ oscillates and does not converge to zero, which violates the necessary condition for convergence of a series. By the Divergence Test, the series must diverge.